

Staphylococcal Bacteremia in HIV Disease

Staphylococcal bacteremia may complicate either primary or secondary cutaneous infections such as psoriasis or scabies. However, the most common source of hematogenous dissemination is an intravenous catheter, accounting for 73 percent of bacteremia cases. As with most infections in HIV disease, staphylococcal infections are frequently recurrent.

Management (see Chap. 177) should include: - Treatment of the acute infection - Management of underlying dermatoses - Eradication of nasal *S. aureus* carriage using mupirocin (pseudomonic acid) ointment - Long-term oral prophylaxis for recurrent infections

Pseudomonas aeruginosa Infections in HIV Disease

Pseudomonas aeruginosa (see Chap. 180) can cause primary cutaneous infections such as cellulitis (e.g., ecthyma gangrenosum), catheter-site infections, and secondary infections of underlying skin conditions like Kaposi sarcoma in advanced HIV disease.

Cutaneous *P. aeruginosa* infection occurs via: - Local invasion through skin breaches - Primary infection of skin appendages in the anogenital or axillary regions - Hematogenous dissemination to the skin

These infections may progress to bacteremia, leading to seeding of multiple organ systems.

Tuberculosis and Other Mycobacterial Infections in HIV Disease

Tuberculosis is the most common opportunistic infection in HIV-infected individuals in developing countries. While cutaneous tuberculosis (see Chap. 184) is relatively uncommon, the incidence of extrapulmonary tuberculosis is higher in HIV-positive individuals: - 15% in non-HIV-infected individuals - 20% to 40% in those with HIV disease - Up to 70% in advanced HIV disease

Non-tuberculous (environmental) mycobacteria reported to cause cutaneous lesions in HIV disease include: - *M. chelonae* - *M. fortuitum* - *M. kansasii* - *M. malmoense* - *M. gordonae* - *M. marinum* - *M. haemophilum*

The most common non-tuberculous mycobacterial pathogen in advanced HIV disease is the *M. avium-intracellulare* complex (MAC). However, MAC rarely presents with cutaneous manifestations. The incidence of MAC infection has declined due to primary prophylaxis with azithromycin and, more recently, effective antiretroviral therapy (HAART).

Bacillary Angiomatosis and Bacillary Peliosis in HIV Disease

Bacillary angiomatosis (BA) and bacillary peliosis (see Chap. 182), caused by *Bartonella henselae* and *B. quintana*, occur predominantly in the setting of advanced HIV-induced immunodeficiency.

Clinical Features of Bacillary Angiomatosis

- Angioproliferative skin lesions resembling: - Cherry hemangiomas - Pyogenic granulomas - Kaposi sarcoma
- Cutaneous lesions are red to violaceous, dome-shaped papules, nodules, or plaques
- Lesion size ranges from a few millimeters to 2–3 cm in diameter
- Dermal vascular lesions with thinned or eroded epidermis
- Subcutaneous nodular lesions may also occur, sometimes lacking the typical red appearance
- Lesions are soft to firm and may be tender
- Number of lesions varies from solitary to over 100; in rare cases, more than 1000

Currently, the prevalence of BA in North America and Western Europe is very low, largely due to improved immune function from HAART and prophylactic strategies targeting opportunistic infections.